

ML Challenge 2025: Smart Product Pricing

Solution Report

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1. Executive Summary

We developed a machine learning pipeline to predict product prices from textual descriptions in the `catalog_content` field. The system combines structured feature extraction (numeric and categorical signals) with text embeddings (TF-IDF + SVD) and a tree-based ensemble regressor. Our final XGBoost model achieved a mean SMAPE of **58.36%** and R^2 of **0.3479**, outperforming the baseline HistGradientBoosting model (SMAPE 60.87%, R^2 0.3402).

2. Methodology Overview

2.1 Problem Analysis

The task was treated as a supervised regression problem, mapping descriptive text data to numerical price outputs. Through exploratory data analysis, patterns such as “*Pack of X*”, “*Value: n Unit: Y*”, and text length statistics showed strong correlation with product price. Hence, both textual and derived numeric representations were combined for modeling.

Key Observations:

- Product quantity and packaging indicators directly influence prices.
- Descriptive length, numerical values, and presence of digits strongly correlate with pricing.
- Pure TF-IDF text models underperform; combining structured and textual features improves robustness.

2.2 Solution Strategy

Approach Type: Single-model regression (tree-based ensemble). **Core Innovation:** Integration of engineered structured features with compressed textual embeddings through a unified pipeline.

Steps followed:

1. Text cleaning, normalization, and regex-based extraction of numerical cues.
2. TF-IDF encoding (word & character n-grams) and dimensionality reduction via SVD.
3. Standardization and encoding of numeric/categorical variables.
4. Model training using RepeatedKFold (5×3) cross-validation.
5. Final model training on the complete dataset and prediction for the test set.

3. Model Architecture

3.1 Architecture Overview

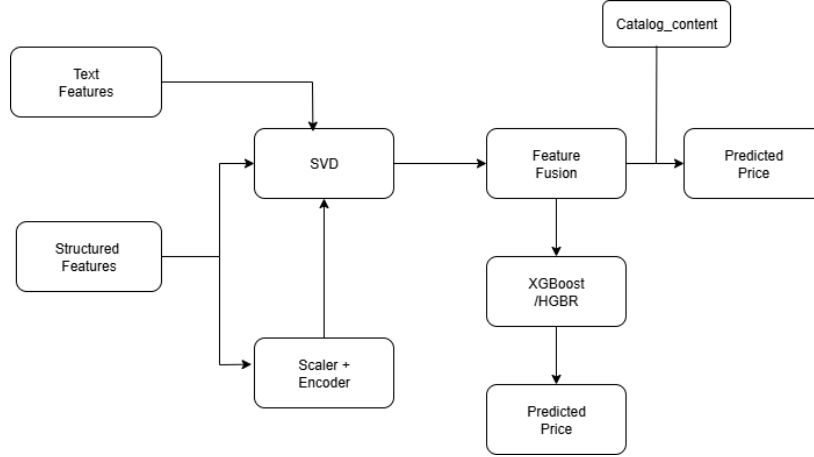


Figure 1: End-to-end model architecture showing feature engineering and regression components.

3.2 Model Components

Text Processing Pipeline:

- Preprocessing: Lowercasing, Unicode normalization, token cleanup.
- Feature extraction: Word-level (1–3 n-grams) and character-level (3–6 n-grams) TF-IDF.
- Dimensionality reduction: TruncatedSVD (300 components, random_state=42).

Structured Features:

- Extracted via regex: *pack_of*, *value_num*, *unit*.
- Derived metrics: *log_pack_of*, *value_per_pack*, *text length*, *digit ratio*, *uppercase ratio*.
- Numeric scaling with StandardScaler; categorical encoding with OneHotEncoder (min_frequency=20).

Model Algorithm:

- **Primary:** XGBoost Regressor (`n_estimators=1200`, `max_depth=8`, `learning_rate=0.05`, `subsample=0.8`, `colsample_bytree=0.8`, `tree_method="hist"`).
- **Fallback:** HistGradientBoostingRegressor (`max_depth=10`, `learning_rate=0.06`, `max_iter=1200`).
- Validation: Repeated K-Fold (5 splits $\times$ 3 repeats) with SMAPE as the main scoring metric.

4. Model Performance

4.1 Validation Results

Model	SMAPE ($\downarrow$)	RMSE ($\downarrow$)	MAE ($\downarrow$)	R ² ($\uparrow$)
HistGradientBoosting	60.8706	27.0295	13.2859	0.3402
XGBoost	58.3671	26.8488	12.5811	0.3479

Table 1: Cross-validation metrics averaged over 15 runs (5 folds $\times$ 3 repeats).

Observations:

- XGBoost consistently outperformed HGBR in SMAPE, MAE, and R².
- Text + structured fusion reduced error by nearly 2.5 SMAPE points.
- Full model training on 75k samples completed in $\sim$ 535 seconds.

5. Conclusion

Our final XGBoost-based pipeline achieved a competitive SMAPE score of 58.36% with robust performance consistency across validation folds. The inclusion of engineered numeric-textual features, along with TF-IDF+SVD embeddings, proved effective for generalization. Future work can explore multi-modal learning by integrating image embeddings from product photos to further improve prediction accuracy.